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NORME DE LA CEI

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IEC STANDARD

Publication 853-1

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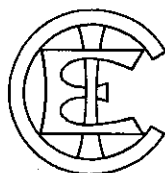
1985

**Calcul des capacités de transport des câbles pour les régimes
de charge cycliques et de surcharge de secours**

**Première partie: Facteur de capacité de transport cyclique pour des câbles
de tensions inférieures ou égales à 18/30 (36) kV**

**Calculation of the cyclic and emergency current
rating of cables**

**Part 1: Cyclic rating factor for cables up to
and including 18/30 (36) kV**



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Bureau Central de la Commission Electrotechnique Internationale
3, rue de Varembe
Genève, Suisse

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**NORME
INTERNATIONALE
INTERNATIONAL
STANDARD**

**CEI
IEC
853-1**

1985

**AMENDEMENT 1
AMENDMENT 1**

1994-09

Amendement 1

**Calcul des capacités de transport des câbles
pour les régimes de charge cycliques et de
surcharge de secours**

Première partie:

**Facteur de capacité de transport cyclique pour
des câbles de tensions inférieures ou égales
à 18/30 (36) kV**

Amendment 1

**Calculation of the cyclic and emergency current
rating of cables**

Part 1:

**Cyclic rating factor for cables up to and
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FOREWORD

This amendment has been prepared by sub-committee 20A: High-voltage cables, of IEC technical committee 20: Electric cables.

The text of this amendment is based on the following documents:

DIS	Report on voting
20A(CO)163	20A(CO)165

Full information on the voting for the approval of this amendment can be found in the report on voting indicated in the above table.

Page 9

2 Symbols

Replace the text concerning the W symbol by the following new text:

W = total joule (current-dependent) losses per cable determined using the steady-state current (according to IEC 287 with a load factor of 100 %) and the conductor resistance at maximum operating temperature.

3 Calculation of cyclic rating factor

Replace, on page 11, the text of the second paragraph by the following new text:

The cyclic rating factor is designated by the letter M , and represents the factor by which the permissible continuous current rating (100 % load factor) can be multiplied to obtain the peak current during a daily cycle, such that during this cycle, the conductor reaches, but does not exceed, the standard maximum permissible temperature.

Page 11

3.1.1 Any load cycle of known shape

Replace, on page 13, after equation (4), the text concerning W by the following new text:

W is the total joule (current-dependent) losses per cable determined using the steady-state current (according to IEC 287, with a load factor of 100 %) and the conductor resistance at maximum operating temperature.

Delete, on page 13, the last sentence of the last paragraph.

Page 23

7 Examples

Add, on page 25, the following text at the end of subclause 7.1, after $M = 1,16$:

If the permissible steady-state current in the cable (100 % load factor), is for instance, $I_R = 450$ A, the maximum current that the cable can carry under peak conditions, for the above shape of load cycle, without exceeding the maximum permissible temperature is:

$$I_{\max} = 1,16 \times 450 = 522 \text{ A}$$

Page 34

Figure 1

Replace the existing figure 1a by the following new figure 1a:

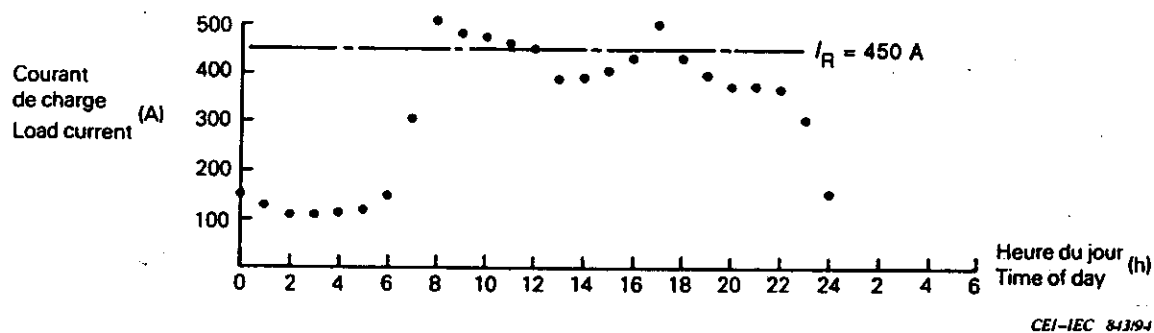


Figure 1a - Cycle de charge

Load cycle

Publications de la CEI préparées par le Comité d'Etudes n° 20

- 55:— Câbles isolés au papier imprégné sous gaine métallique pour des tensions assignées inférieures ou égales à 18/30 kV (avec âmes conductrices en cuivre ou aluminium et à l'exclusion des câbles à pression de gaz et à huile fluide).
- 55-1 (1978) Première partie: Essais.
Modification n° 1 (1989).
- 55-2 (1981) Deuxième partie: Généralités et exigences de construction.
Modification n° 1 (1989).
- 141:— Essais de câbles à huile fluide, à pression de gaz et de leurs dispositifs accessoires.
- 141-1 (1993) Première partie: Câbles au papier à huile fluide et à gaine métallique et accessoires pour des tensions alternatives inférieures ou égales à 400 kV.
- 141-2 (1963) Deuxième partie: Câbles à pression de gaz interne et accessoires pour des tensions alternatives inférieures ou égales à 275 kV.
Modification n° 1 (1967).
- 141-3 (1963) Troisième partie: Câbles à pression de gaz externe (à compression de gaz) et accessoires pour des tensions alternatives inférieures ou égales à 275 kV.
Modification n° 1 (1967).
- 141-4 (1980) Quatrième partie: Câbles à huile fluide en tuyau à isolation de papier imprégné sous forte pression d'huile et accessoires pour des tensions alternatives inférieures ou égales à 400 kV.
Amendement n° 1 (1990).
- 173 (1964) Couleurs pour les conducteurs des câbles souples.
- 183 (1984) Guide pour le choix des câbles à haute tension.
Amendement n° 1 (1990).
- 227:— Conducteurs et câbles isolés au polychlorure de vinyle, de tension nominale au plus égale à 450/750 V.
- 227-1 (1993) Partie 1: Prescriptions générales.
- 227-2 (1979) Deuxième partie: Méthodes d'essais.
Modification n° 1 (1985).
- 227-3 (1993) Partie 3: Conducteurs pour installations fixes.
- 227-4 (1992) Partie 4: Câbles sous gaine pour installations fixes.
- 227-5 (1979) Cinquième partie: Câbles souples.
Modification n° 1 (1987).
- 227-6 (1985) Sixième partie: Câbles pour ascenseurs et câbles pour connexions souples.
- 228 (1978) Ames des câbles isolés. Guide pour les limites dimensionnelles des âmes circulaires.
Amendement 1 (1993).
- 228A (1982) Premier complément.
- 229 (1982) Essais sur les gaines extérieures des câbles, qui ont une fonction spéciale de protection et sont appliquées par extrusion.
- 230 (1966) Essais de choc des câbles et de leurs accessoires.
- 245:— Conducteurs et câbles isolés au caoutchouc – Tension assignée au plus égale à 450/750 V.
- 245-1 (1994) Partie 1: Prescriptions générales.
- 245-2 (1994) Partie 2: Méthodes d'essais.
- 245-3 (1994) Partie 3: Conducteurs isolés au silicone, résistant à la chaleur.
- 245-4 (1994) Partie 4: Câbles souples.
- 245-5 (1994) Partie 5: Câbles pour ascenseurs.

(suite)

IEC publications prepared by Technical Committee No. 20

- 55:— Paper-insulated metal-sheathed cables for rated voltages up to 18/30 kV (with copper or aluminium conductors and excluding gas-pressure and oil-filled cables).
- 55-1 (1978) Part 1: Tests.
Amendment No. 1 (1989).
- 55-2 (1981) Part 2: General and construction requirements.
Amendment No. 1 (1989).
- 141:— Tests on oil-filled and gas-pressure cables and their accessories.
- 141-1 (1993) Part 1: Oil-filled, paper-insulated, metal-sheathed cables and accessories for alternating voltages up to and including 400 kV.
- 141-2 (1963) Part 2: Internal gas-pressure cables and accessories for alternating voltages up to 275 kV.
Amendment No. 1 (1967).
- 141-3 (1963) Part 3: External gas-pressure (gas compression) cables and accessories for alternating voltages up to 275 kV.
Amendment No. 1 (1967).
- 141-4 (1980) Part 4: Oil-impregnated paper-insulated high-pressure oil-filled pipe-type cables and accessories for alternating voltages up to and including 400 kV.
Amendment No. 1 (1990).
- 173 (1964) Colours of the cores of flexible cables and cords.
- 183 (1984) Guide to the selection of high-voltage cables.
Amendment No. 1 (1990).
- 227:— Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V.
- 227-1 (1993) Part 1: General requirements.
- 227-2 (1979) Part 2: Test methods.
Amendment No. 1 (1985).
- 227-3 (1993) Part 3: Non-sheathed cables for fixed wiring.
- 227-4 (1992) Part 4: Sheathed cables for fixed wiring.
- 227-5 (1979) Part 5: Flexible cables (cords).
Amendment No. 1 (1987).
- 227-6 (1985) Part 6: Lift cables and cables for flexible connections.
- 228 (1978) Conductors of insulated cables. Guide to the dimensional limits of circular conductors.
Amendment 1 (1993).
- 228A (1982) First supplement.
- 229 (1982) Tests on cable oversheaths which have a special protective function and are applied by extrusion.
- 230 (1966) Impulse tests on cables and their accessories.
- 245:— Rubber insulated cables – Rated voltages up to and including 450/750 V.
- 245-1 (1994) Part 1: General requirements.
- 245-2 (1994) Part 2: Test methods.
- 245-3 (1994) Part 3: Heat resistant silicone insulated cables.
- 245-4 (1994) Part 4: Cords and flexible cables.
- 245-5 (1994) Part 5: Lift cables.

(suite)

**Publications de la CEI préparées
par le Comité d'Études n° 20 (suite)**

- 811-1-1 (1993) Section un: Mesure des épaisseurs et des dimensions extérieures – Détermination des propriétés mécaniques.
- 811-1-2 (1985) Section deux: Méthodes de vieillissement thermique. Modification n° 1 (1989).
- 811-1-3 (1993) Section 3: Méthodes de détermination de la masse volumique – Essais d'absorption d'eau – Essai de rétraction.
- 811-1-4 (1985) Section quatre: Essais à basse température. Amendement 1 (1993).
- 811-2:– Deuxième partie: Méthodes spécifiques pour les mélanges élastomères.
- 811-2-1 (1986) Section un: Essai de résistance à l'ozone – Essai d'allongement à chaud – Essai de résistance à l'huile. Amendement 1 (1992). Amendement 2 (1993).
- 811-3:– Troisième partie: Méthodes spécifiques pour les mélanges PVC.
- 811-3-1 (1985) Section un: Essai de pression à température élevée – Essais de résistance à la fissuration. Amendement 1 (1993).
- 811-3-2 (1985) Section deux: Essai de perte de masse – Essai de stabilité thermique. Amendement 1 (1993).
- 811-4:– Quatrième partie: Méthodes spécifiques pour les mélanges polyéthylène et polypropylène.
- 811-4-1 (1985) Section un: Résistance aux craquelures sous contraintes dues à l'environnement – Essai d'enroulement après vieillissement thermique dans l'air – Mesure de l'indice de fluidité à chaud – Mesure dans le PE du taux de noir de carbone et/ou des charges minérales. Modification n° 1 (1988). Amendement 2 (1993).
- 811-4-2 (1990) Section deux: Allongement à la rupture après pré-conditionnement – Essai d'enroulement après pré-conditionnement – Essai d'enroulement après vieillissement thermique dans l'air – Mesure de l'augmentation de masse – Essai de stabilité à long terme (annexe A) – Méthode d'essai pour l'oxydation catalytique par le cuivre (annexe B).
- 811-5-1 (1990) Cinquième partie: Méthodes spécifiques pour les matières de remplissage – Section un: Point de goutte – Séparation d'huile – Fragilité à basse température – Indice d'acide total – Absence de composés corrosifs – Permittivité à 23 °C – Résistivité en courant continu à 23 °C et 100 °C.
- 840 (1988) Essais des câbles de transport d'énergie à isolation extrudée pour des tensions assignées supérieures à 30 kV ($U_m = 36$ kV) et jusqu'à 150 kV ($U_m = 170$ kV). Amendement 2 (1993).
- 853:– Calcul des capacités de transport des câbles pour les régimes de charge cycliques et de surcharge de secours.
- 853-1 (1985) Première partie: Facteurs de capacité de transport cyclique pour des câbles de tensions inférieures ou égales à 18/30 (36) kV. Amendement 1 (1994).

(suite)

**IEC publications prepared
by Technical Committee No. 20 (continued)**

- 811-1-1 (1993) Section One: Measurement of thickness and overall dimensions – Tests for determining the mechanical properties.
- 811-1-2 (1985) Section Two: Thermal ageing methods. Amendment No. 1 (1989).
- 811-1-3 (1993) Section 3: Methods for determining the density – Water absorption tests – Shrinkage test.
- 811-1-4 (1985) Section Four: Tests at low temperature. Amendment 1 (1993).
- 811-2:– Part 2: Methods specific to elastomeric compounds.
- 811-2-1 (1986) Section One: Ozone resistance test – Hot set test – Mineral oil immersion test. Amendment 1 (1992). Amendment 2 (1993).
- 811-3:– Part 3: Methods specific to PVC compounds.
- 811-3-1 (1985) Section One: Pressure test at high temperature – Tests for resistance to cracking. Amendment 1 (1993).
- 811-3-2 (1985) Section Two: Loss of mass test – Thermal stability test. Amendment 1 (1993).
- 811-4:– Part 4: Methods specific to polyethylene and polypropylene compounds.
- 811-4-1 (1985) Section One: Resistance to environmental stress cracking – Wrapping test after thermal ageing in air – Measurement of the melt flow index – Carbon black and/or mineral content measurement in PE. Amendment No. 1 (1988). Amendment 2 (1993).
- 811-4-2 (1990) Section Two: Elongation at break after pre-conditioning – Wrapping test after pre-conditioning – Wrapping test after thermal ageing in air – Measurement of mass increase – Long-term stability test (Appendix A) – Test method for copper-catalysed oxidative degradation (Appendix B).
- 811-5-1 (1990) Part 5: Methods specific to filling compounds – Section One: Drop point – Separation of oil – Lower temperature brittleness – Total acid number – Absence of corrosive components – Permittivity at 23 °C – D.C. resistivity at 23 °C and 100 °C.
- 840 (1988) Tests for power cables with extruded insulation for rated voltages above 30 kV ($U_m = 36$ kV) up to 150 kV ($U_m = 170$ kV). Amendment 2 (1993).
- 853:– Calculation of the cyclic and emergency current rating of cables.
- 853-1 (1985) Part 1: Cyclic rating factor for cables up to and including 18/30 (36) kV. Amendment 1 (1994).

(continued)

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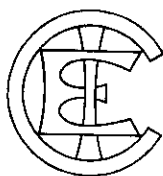
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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**CALCULATION OF THE CYCLIC AND EMERGENCY CURRENT RATING
OF CABLES****Part 1: Cyclic rating factor for cables up to and including 18/30 (36) kV**

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendation and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This standard has been prepared by Sub-Committee 20A: High-voltage Cables, of IEC Technical Committee No. 20: Electric Cables.

The text of this standard is based on the following documents:

Six Months' Rule	Report on Voting
20A(CO)93	20A(CO)97

Further information can be found in the Report on Voting indicated in the table above.

The following IEC publication is quoted in this standard:

Publication No. 287 (1982): Calculation of the Continuous Current Rating of Cables (100% Load Factor).

CALCULATION OF THE CYCLIC AND EMERGENCY CURRENT RATING OF CABLES

Part 1: Cyclic rating factor for cables up to and including 18/30 (36) kV

INTRODUCTION

This standard gives methods for calculating the cyclic rating factor for cables where the internal thermal capacitance can be neglected. The method is based on the simplified procedure published in a CIGRÉ document.*

It is intended to issue a second part detailing the full method (a brief résumé of which is set out in Appendix A) which will provide for those cables where the thermal capacitance should be included. This second part will also include methods for the calculation of emergency ratings.

The simplified methods in this standard require only a knowledge of the full shape of the load variation for not more than 6 h immediately preceding the time of maximum temperature and an average value for times before that. This reduces the amount of work involved, but places some restrictions on the type of cable and shape of load cycle that can be dealt with. The methods can be applied to all sizes and types of cable for nominal voltages up to and including 18/30 (36) kV, with any shape of load cycle, when the margin of error is not expected to exceed 5% on the safe side. Criteria for limited application of these simplified methods to cables for higher voltages are given in Clause 6.

Change in conductor resistance with temperature is neglected. This can be shown to result in a small error on the safe side.

1. Scope

No account has been taken at present of possible effects of moisture migration on the thermal resistivity of the soil. This is in line with the approach already adopted for the calculation of steady-state ratings in uniform soil as given in IEC Publication 287: Calculation of the Continuous Current Rating of Cables (100% Load Factor). Methods for dealing with regions having different thermal characteristics and with drying out of the soil are under consideration.

The methods in this standard apply to cables buried in uniform soil, either directly or in ducts, when carrying a load which varies cyclically over a 24 h period, the shape of each daily cycle being substantially the same. It is assumed that the voltage has been applied for a sufficiently long time for the conductor temperature rise due to dielectric loss to have reached a steady state. The total temperature rise of the conductor above ambient is then the sum of the steady-state temperature rise due to the dielectric loss (as given by IEC Publication 287) and the transient temperature variations due to changes of current.

* Current Ratings of Cables for Cyclic and Emergency Loads, Part 1: Electra No. 24, October 1972, Page 63.

For cables in air the conductor temperature follows changes in load current sufficiently rapidly so that the usual daily cycles do not permit peak loads greater than the steady state value.

Numerical values are provided for the evaluation of cyclic rating factors for cables buried at 1 m depth in soil having a diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$, but methods for dealing with other conditions are included. It is assumed that soil properties are constant in both time and space.

2. Symbols

Symbol	Definition	Unit
B	= quality used in Clause 3 and Appendix A	
D_e	= external diameter of a cable or a duct	m
F	= coefficient used to express the steady state mutual heating caused by other cables in a group	
I	= value of load current observed at hourly intervals during a 24 h cycle	A
$I_{\max}$	= highest of the hourly values of load current	A
L	= depth of laying measured to cable axis	m
M	= rating factor due to a cyclic variation of load current	
T_4	= external thermal resistance of a single cable or duct either in isolation or as one of a group	K.m/W
ΔT_4	= increase in external thermal resistance of the cable under consideration due to the presence of other cables in the group	K.m/W
W	= total joule (current dependent) losses per cable determined using conductor resistance at maximum operating temperature	W/m
$Y_0, Y_1, Y_2, \dots, Y_i$	= coefficient proportional to the current-dependent losses in a cable between (i) and $(i+1)$ hours prior to the instant of highest conductor temperature	
i	= time prior to expected time of maximum temperature	h
k	= ratio of cable (or duct) external surface temperature rise above ambient to conductor temperature rise above ambient under steady-state conditions	
k_1	= value of k for a cable in a group	
p	= fraction of daily load cycle during which the load is not at its maximum value	
t	= time prior to expected time of maximum temperature	s
$\theta(\infty)$	= conductor steady state temperature rise above ambient (but see Sub-clause 3.1.1 when dielectric loss is not negligible)	K
$\theta_R(i)$	= conductor temperature rise above ambient at time i hours due to the application of a step function of load current equal to the sustained (100% load factor) rated current. $\theta_R(\infty)$ is the steady state conductor temperature rise due to the sustained rated current (see also remarks to $\theta(\infty)$ when dielectric loss is not negligible)	K
$\Phi_0, \Phi_1, \Phi_2, \dots, \Phi_i$	= function defined in Appendix A. Numerical values given in Table II	
$\alpha(i)$	= cable conductor temperature attainment factor at time (i) hours, $\alpha(\infty) = 1$	
$\beta(i)$	= cable (or duct) outer surface temperature attainment factor at time (i) hours, $\beta(\infty) = 1$	
$\gamma(i)$	= function used in calculating cyclic rating factors for one of a group of cables	
δ	= soil thermal diffusivity	m^2/s
μ	= loss load factor for load current cycle under consideration	
ϱ_T	= soil thermal resistivity	K.m/W

3. Calculation of cyclic rating factor

The method varies depending on the shape and the amount of detail known for the load cycle. The basic method is given for reference in Appendix A, and the formulae given here are simplifications which can be made according to the case being dealt with. Table I provides a guide to the method to be selected for each case.

TABLE I
Guide to selection of methods

Description of circuit	Description of load cycle	Sub-clause
Isolated cable or Isolated circuit of three single-core cables Direct buried or in ducts. Cables or ducts touching each other	Any type of load variation	3.1.1
	Sustained maximum for at least 6 h, constant fraction for remainder	3.1.2
	Sustained maximum for at least 6 h, any type of variation for remainder	3.1.3
	Nothing known about load cycle except the loss load factor μ	3.1.4
Group of cables with equal losses Direct buried or in ducts. Cables or ducts not touching	Any type of load variation	3.2.1
	Sustained maximum for at least 6 h, any type of variation for remainder	3.2.2
Group of circuits, each circuit having three cables, all cables having equal losses. Direct buried or in ducts. Cables or ducts in each circuit touching, circuits are spaced	Any type of load variation	3.3
	Sustained maximum for at least 6 h, any type of variation for remainder	3.3

The cyclic rating factor is denoted by the letter M , and is that factor by which a daily cyclic current, whose maximum value is equal to the sustained (100% load factor) rated current permissible under steady-state conditions, may be multiplied for the conductor to attain, but not exceed, the standard maximum permissible temperature.

3.1 Single isolated circuit of:

- One multicore cable, direct buried or in a duct.
- Three cables direct buried, in trefoil touching or flat touching formation.
- Three cables in ducts, in trefoil touching or flat touching formation.

3.1.1 Any load cycle of known shape

The daily load cycle is first expressed as 24 hourly values by scaling the whole cycle so that its maximum value is equal to unity (see Figures 1a and 1b, page 34). The magnitude of each hourly value is then squared to give 24 values representing the cycle of cable joule losses (see Figure 1c, page 34). This curve is then expressed as a series of step function changes so as to give a stepped curve having approximately the same average value as the original one (see Figure 1d, page 34). Denote the ordinates of the stepped curve for the six hourly periods before the occurrence of the highest temperature by $Y_0, Y_1, Y_2, \dots, Y_5$. It should be noted that the particular way in which the load and loss-load curves of Figure 1 have been drawn assumes that the measured values as given in the example of Sub-clause 7.1 apply to periods of time which extend half an hour before and after each observation time. Location of the time of the highest temperature is done by inspection, bearing in mind that although it

usually occurs at the end of the period of highest current, this may not always be the case. The temperature also depends on the general level of current preceding that period. Thus, in the example given in Sub-clause 7.1 and Figure 1, the highest temperature occurs at 17.5 h although the highest value of current was at 8 h.

The loss load factor (μ) is given by:

$$\mu = \frac{1}{24} \sum_{i=0}^{23} Y_i \quad (1)$$

The cyclic factor (M) is given by:

$$M = \frac{1}{((1-k)Y_0 + k\{B + \mu[1 - \beta(6)]\})^{1/2}} \quad (2)$$

where:

$$B = Y_0\Phi_0 + Y_1\Phi_1 + Y_2\Phi_2 + Y_3\Phi_3 + Y_4\Phi_4 + Y_5\Phi_5 \quad (3)$$

Values of Φ_0 to Φ_5 and of $[1 - \beta(6)]$ for a cable at a depth of 1 m and a soil thermal diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$ are given in Table IV. These values may be used for cables or ducts at depths within the range 0.75 m to 1.5 m with an error generally less than 1.5% in the factor M . Methods for dealing with any other values of depth and soil thermal diffusivity are given in Clause 4.

$$k = \frac{WT_4}{\theta(\infty)} \quad (4)$$

where:

W are the total joule losses per cable at maximum operating temperature, W/m

T_4 is the external thermal resistance of single cable or duct, K.m/W

$\theta(\infty)$ is the conductor steady-state temperature rise above ambient, K

If the dielectric losses are not negligible then $\theta(\infty)$ is defined as the conductor temperature rise in the steady state which is due to joule losses only, i.e. to the losses W per unit length of cable as calculated in IEC Publication 287.

The external thermal resistance (T_4) is given by:

i) For a single isolated cable or duct

$$T_4 = \frac{\rho_T}{2\pi} \ln \left(\frac{4L}{D_e} \right) \quad (5)$$

where:

ρ_T is the soil thermal resistivity, K.m/W

L is the depth of laying, metres

D_e is the external diameter of cable or duct, metres

ii) For isolated circuits of three cables or ducts touching each other, either in trefoil or flat formation.

Formulae for T_4 should be taken from IEC Publication 287.

The value of cable or duct diameter to be used for obtaining Φ_0 to Φ_5 and $[1 - \beta(6)]$ is that of a single cable. This treatment assumes that a group of cables touching each other acts thermally as one unit.

3.1.2 *Flat topped load cycle consisting of a sustained maximum current for at least 6 h and a constant fraction of that current for the rest of the cycle*

If the duration of the sustained maximum load is x hours ($x \geq 6$ h) and the duration of the constant fraction of the maximum load for the remaining $(24-x)$ hours is p , then:

$$M = \frac{1}{\left[1 - k(1-p^2) \left(\frac{24-x}{24}\right) [1-\beta(6)]\right]^{1/2}} \quad (6)$$

3.1.3 *Flat topped load cycle having a sustained maximum current for at least 6 h and no restriction on the shape for the remainder of the cycle, except that the maximum temperature rise occurs at the end of the sustained maximum current*

$$M = \frac{1}{(1-k(1-\mu)[1-\beta(6)])^{1/2}} \quad (7)$$

3.1.4 *Load cycle where the shape is unknown and only the loss-load factor μ is known*

For this case it can be shown that the cycle may be considered as flat topped and equation (7) may be used.

3.2 *Group of cables with equal losses, cables or ducts not touching*

3.2.1 *Any shape of load cycle*

$$M = \frac{1}{\left(\left\{\sum_{i=0}^5 Y_i \left[\frac{\theta_R(i+1)}{\theta_R(\infty)} - \frac{\theta_R(i)}{\theta_R(\infty)}\right]\right\} + \mu \left[1 - \frac{\theta_R(6)}{\theta_R(\infty)}\right]\right)^{1/2}} \quad (8)$$

where:

$$\theta_R(0) = 0$$

$$\frac{\theta_R(i)}{\theta_R(\infty)} = 1 - k_1 + k_1 \gamma(i) \quad (9)$$

$$\gamma(i) = \frac{-Ei\left(-\frac{D_e^2}{16t\delta}\right) + (N-1)\left\{-Ei\left(-\frac{d_i^2}{16t\delta}\right)\right\}}{2 \ln\left(\frac{4LF}{D_e}\right)} \quad (10)$$

$$t = 3600 i$$

δ is the soil thermal diffusivity

$-Ei(-x)$ is the exponential integral function. Evaluation of this formula is facilitated by using the nomogram for exponential integral provided in Figures 3 and 4, pages 36 and 37.

Note. – Image terms are not required in this formula as the time periods involved are only up to 6 h.

$$F = \frac{d'_{p1} \cdot d'_{p2} \cdots d'_{pk} \cdots d'_{p(N-1)}}{d_{p1} \cdot d_{p2} \cdots d_{pk} \cdots d_{p(N-1)}} \quad (11)$$

there are $(N-1)$ terms with the term $\frac{d'_{pp}}{d_{pp}}$ excluded.

$$d_i = \frac{4L}{F^{1/(N-1)}} \quad (12)$$

Distances d_{pk} and d'_{pk} are defined in Figure 2, page 35. The values of D_e and L to be used for calculating $\gamma(i)$ refer to the cable or duct containing the hottest cable of the group.

$$k_1 = \frac{W(T_4 + \Delta T_4)}{\theta(\infty)} \quad (13)$$

T_4 is the external thermal resistance of the hottest cable in the group regarded as an isolated cable, and

$$\Delta T_4 = \frac{\theta_T \ln F}{2\pi} \quad (14)$$

k_1 is the ratio of cable or duct outside surface temperature rise above ambient to the conductor temperature rise above ambient under steady state conditions with all cables of the group equally loaded and each having a total heat loss of W per unit length.

3.2.2 *Flat topped load cycle having a sustained maximum current for at least 6 h, no restriction on the shape of the rest of the cycle, except that the maximum temperature rise occurs at the end of the sustained maximum current*

$$M = \frac{1}{\left(\frac{\theta_R(6)}{\theta_R(\infty)} + \mu \left\{ 1 - \frac{\theta_R(6)}{\theta_R(\infty)} \right\} \right)^{1/2}} \quad (15)$$

3.3 *Group of circuits of three equally loaded, identical, touching cables or ducts having equal losses*

The methods are the same as those given in Sub-clause 3.2 above, except that

- i) The external thermal resistance T_4 , on which the value of k_1 is dependent, relates to an isolated circuit of three touching cables or ducts (see Sub-clause 3.1.1).
- ii) The value of N to be used for the calculation of $\gamma(i)$, F and d_i in equations (10), (11) and (12) is the number of circuits (of three single-core cables per circuit) and the values of d'_{pk} and d_{pk} are given by:

d_{pk} = distance from the centre of circuit k to centre of circuit containing the hottest cable,

d'_{pk} = distance of image of centre of circuit k to centre of circuit containing the hottest cable.

- iii) The right-hand side of equation (14) should be multiplied by 3, with F given by Item ii) above.

iv) W is the total loss per unit length per single cable or duct.

- v) The value of the cable or duct diameter to be assumed for the calculation of $\gamma(i)$ is that of an isolated cable or duct.

4. Method of dealing with different soil resistivities, soil thermal diffusivities and depths of laying

The quantities Φ_0 to Φ_5 , $[1 - \beta(6)]$, k and k_1 are dependent on soil properties and depth of laying. The tabulated values (see Table IV) can be used over a popular range of conditions (see Sub-clause 3.1.1), but it may be necessary to compute them for conditions outside this range.

- i) Numerical values for Φ_0 to Φ_5 , and $[1 - \beta(6)]$ for any specified value of depth and soil thermal diffusivity may be obtained from:

$$\Phi_0 = \beta(1) \text{ and } \Phi_i = [\beta(i+1) - \beta(i)] \text{ for } i = 1, 2 \dots 5$$

$$\beta(i) = \frac{-Ei \left(-\frac{D_e^2}{16i\delta} \right)}{2 \ln (4L/D_e)} \text{ where } t = 3600 i \quad (16)$$

- ii) The factors k and k_1 are dependent on depth of laying and soil resistivity. If the laying conditions "a", for which a value of k is known, change to conditions "b" then the new value of k is given by:

$$k(b) = \frac{1}{1 + \frac{T_4}{T'_4} \left(\frac{1 - k(a)}{k(a)} \right)} \quad (17)$$

where T_4 and T'_4 are the external thermal resistances for conditions "a" and "b" respectively. If the depth is not changed then $T_4/T'_4 = \rho(a)/\rho(b)$.

A new value of k_1 is obtained in the same way.

5. Soil thermal diffusivity

For calculation of cyclic rating factors it is not necessary to know accurately the value of thermal diffusivity and it is generally satisfactory to use the tabulated functions given in this standard which are based on the value of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$, and correspond roughly to a soil having a thermal resistivity of 1 K.m/W and a moisture content of around 7% of dry mass. For exceptional cases and where moisture content is likely to be much lower it is advisable to use the methods given below to calculate alternative values.

- i) The thermal diffusivity δ can be estimated if the following quantities have been measured: density, moisture content by mass and thermal resistivity.

$$\delta = \frac{10^{-3}}{\rho_T d (0.82 + 0.042 \eta)} \text{ m}^2/\text{s} \quad (18)$$

where:

ρ_T is the thermal resistivity, K.m/W

d is the dry density, kg/m^3

η is the moisture content, % of dry mass.

Alternatively there are well known laboratory methods for measuring thermal diffusivity.

- ii) If only the thermal resistivity is known, then the thermal diffusivity can be obtained from Table II.

TABLE II
Values of soil thermal diffusivity

Thermal resistivity (K.m/W)	Thermal diffusivity (m ² /s)
0.5	0.8×10^{-6}
0.6	0.7×10^{-6}
0.7	0.6×10^{-6}
0.8	0.6×10^{-6}
0.9	0.5×10^{-6}
1.0	0.5×10^{-6}
1.2	0.4×10^{-6}
1.5	0.4×10^{-6}
2.0	0.3×10^{-6}
2.5	0.2×10^{-6}
3.0	0.2×10^{-6}

Note. – These were obtained from an empirical relationship for homogeneous and moist soils.

- iii) If no data are available on the soil, use a value of thermal diffusivity of 0.5×10^{-6} m²/s.

6. Criteria for application of methods to cables of higher voltage

The magnitude of error, which will be on the safe side, involved in neglecting cable thermal capacitance* depends on the shape of the load cycle, which for convenience can be represented by the loss-load factor μ . In general, the error in cyclic rating factor calculated by methods given in the original CIGRÉ document was estimated not to exceed $\pm 3\%$ provided that the effect of cable thermal capacitance was included and μ was not less than 0.4.

In cases where μ is not less than 0.4 and thermal capacitance is neglected there may be extreme cases where the margin will exceed 5% on the safe side.

It has been estimated that cable thermal capacitance can be neglected for:

- All sizes and types of cable for nominal voltages up to and including 18/30 (36) kV for any shape of load cycle. The margin is estimated not to exceed 5% on the safe side.
- All cables where the loss-load factor μ is not less than 0.65, the average of the ordinates Y_0 , Y_1 and Y_2 (see Appendix A) is not less than 0.9 and the average of Y_3 , Y_4 and Y_5 is not less than 0.7. The margin is estimated to be not greater than 3% on the safe side for self-contained cables and 5% for pipe-type cables.
- Self-contained cables where the loss-load factor is not less than 0.4 and the ordinates Y_0 to Y_5 are as in Item b) above. The margin is estimated not to exceed 5% on the safe side.

* When cable thermal capacitance is neglected $\alpha(i) = 1$, see Appendix A, equation 23.

7. Examples

- 7.1 A single multicore cable having an external diameter of 0.06 m directly buried at a depth of 1 m in soil having a diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$ and a resistivity of 1.2 K.m/W. The maximum conductor temperature is 70°C and the ground temperature is 15°C. A power loss of 48.13 W/m is assumed.

The load cycle is given by the values of current indicated in Table III.

TABLE III

Example

Time (h)	Current (A)	Cyclic load/highest load	Value of Y_i	Y_i
0	151	0.302	0.091	Y_{17}
1	124	0.247	0.061	Y_{16}
2	114	0.227	0.052	Y_{15}
3	116	0.232	0.054	Y_{14}
4	118	0.235	0.056	Y_{13}
5	123	0.246	0.061	Y_{12}
6	145	0.290	0.084	Y_{11}
7	300	0.600	0.360	Y_{10}
8	500	1.000	1.000	Y_9
9	475	0.950	0.902	Y_8
10	470	0.940	0.884	Y_7
11	455	0.910	0.828	Y_6
12	446	0.892	<u>0.796</u>	Y_5
13	385	0.770	<u>0.593</u>	Y_4
14	386	0.772	<u>0.596</u>	Y_3
15	400	0.800	<u>0.640</u>	Y_2
16	427	0.853	<u>0.728</u>	Y_1
17	498	0.996	<u>0.992</u>	Y_0
18	427	0.853	0.728	Y_{23}
19	395	0.790	0.624	Y_{22}
20	370	0.740	0.548	Y_{21}
21	370	0.740	0.548	Y_{20}
22	361	0.722	0.521	Y_{19}
23	300	0.600	0.360	Y_{18}

These are plotted in Figures 1a to 1c, page 34.

$$\mu = \frac{0.091 + 0.061 + 0.052 + \dots + 0.360}{24} = 0.504, \text{ from equation (1)}$$

The equivalent stepped cycle is shown in Figure 1d, page 34, having ordinates given in the last column (Y_i) of the example (Table III).

The whole of the stepped curve has been drawn in for the purpose of illustration of the example only. The ordinates required to calculate the cyclic rating factor are underlined in Table III and shown as heavy lines in Figure 1d. The expected time of occurrence of the highest temperature is 17.5 h.

Taking Y_i from Table III and Φ_i from Table IV.

Y_i	Φ_i	$(Y_i \Phi_i)$
Y_0 0.992	0.193	0.1915
Y_1 0.728	0.076	0.0553
Y_2 0.640	0.046	0.0294
Y_3 0.596	0.033	0.0197
Y_4 0.593	0.026	0.0154
Y_5 0.796	0.021	0.0167
$B = \sum Y_i \Phi_i =$		0.3280

B = 0.328, from equation (3)
 T_a = 0.8 K.m/W, from equation (5)
 W = 48.13 W/m

$\theta_R(\infty)$ = 55 K

k = 0.7, from equation (4)
 $[1 - \beta(6)]$ = 0.605, from Table IV

Thus from equation (2):

$$M = \frac{(1 - 0.7) 0.992 + 0.7 \{0.328 + 0.504 (0.605)\}^{1/2}}{1}$$

$$M = 1.16$$

7.2

Three multicore cables having an external diameter of 0.05 m laid spaced 0.2 m apart axially at a depth of 1 m in soil having a diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$. The maximum conductor temperature is 70°C and the ground temperature is 15°C. A power loss of 27.5 W/m per cable in a group of three cables is assumed. The load cycle, which is the same for all cables, has a constant value for the 6 h preceding time of highest temperature; μ for the whole cycle is 0.6. The calculation is performed for the centre cable.

From Figure 2

$$d'_{p1} = d'_{p3} = \sqrt{(2^2 + 0.2^2)} = 2.01 \text{ m}$$

$$d_{p1} = d_{p3} = 0.2 \text{ m}$$

$$F = \frac{2.01 \times 2.01}{0.2 \times 0.2} = 101 \quad N = 3 \quad \text{(from equation (11))}$$

$$d'_1 = \frac{4 \times 1}{(101)^{1/2}} = 0.398 \text{ m} \quad \text{(from equation (12))}$$

$\gamma(6)$ is obtained from equation (10), with $t = 6 \text{ h}$ ($t = 21600 \text{ s}$)

$$t\delta = 21600 \times 0.5 \times 10^{-6} = 0.0108 \text{ m}^2$$

$$16t\delta = 0.1728 \text{ m}^2$$

$$\frac{16t\delta}{D^2} = \frac{0.1728}{0.05^2} = 0.01447; -Ei(-x) = 3.67 \quad \text{(see Figure 3)}$$

$$\frac{16t\delta}{d'^2} = \frac{0.1728}{0.398^2} = 0.9167; -Ei(-x) = 0.25 \quad \text{(see Figure 4)}$$

$$\gamma(6) = \frac{17.994}{3.67 + 2(0.25)} = 0.232 \quad \text{(from equation (10))}$$

Rating data:

$$\begin{aligned}
 \varrho_T &= 1.2 \text{ K.m/W} \\
 T_4 &= 0.837 \text{ K.m/W (for a single isolated cable)} \quad (\text{from equation (5)}) \\
 W &= 27.5 \text{ W/m} \\
 \theta_R(\infty) &= 55 \text{ K}
 \end{aligned}$$

For a group of three cables:

$$\Delta T_4 = \frac{1.2 \ln(101)}{2\pi} = 0.881 \text{ K.m/W} \quad (\text{From equation (14)})$$

$$k_1 = \frac{27.5 (0.837 + 0.881)}{55} = 0.86 \quad (\text{from equation (13)})$$

$$\frac{\theta_R(6)}{\theta_R(\infty)} = 1 - 0.86 + 0.86(0.232) = 0.3395 \quad (\text{from equation (9)})$$

Thus from equation (15):

$$M = \frac{1}{(0.3395 + 0.6[1 - 0.3395])^{1/2}}$$

$$M = 1.17$$

TABLE IV

Values of Φ_i and $1 - \beta(6)$ for soil thermal diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$

	Cable diameter or outer duct diameter (m)														
	0.030	0.035	0.040	0.045	0.050	0.055	0.060	0.065	0.070	0.075	0.080	0.085	0.090	0.095	0.100
Φ_0	0.298	0.277	0.257	0.239	0.223	0.207	0.193	0.180	0.168	0.156	0.145	0.135	0.126	0.117	0.100
Φ_1	0.069	0.071	0.072	0.073	0.074	0.075	0.076	0.076	0.076	0.076	0.075	0.074	0.074	0.074	0.073
Φ_2	0.041	0.042	0.043	0.044	0.045	0.045	0.046	0.046	0.047	0.047	0.047	0.048	0.048	0.048	0.048
Φ_3	0.029	0.030	0.031	0.031	0.032	0.033	0.033	0.034	0.034	0.034	0.035	0.035	0.035	0.035	0.035
Φ_4	0.023	0.023	0.024	0.025	0.025	0.025	0.026	0.026	0.026	0.027	0.027	0.027	0.027	0.027	0.028
Φ_5	0.019	0.019	0.020	0.020	0.020	0.021	0.021	0.021	0.022	0.022	0.022	0.023	0.023	0.023	0.023
$1 - \beta(6)$	0.521	0.538	0.553	0.568	0.581	0.594	0.605	0.617	0.627	0.638	0.648	0.657	0.667	0.676	0.684

	Cable diameter or outer duct diameter (m)														
	0.105	0.110	0.115	0.120	0.125	0.130	0.135	0.140	0.145	0.150	0.155	0.160	0.165	0.170	0.175
Φ_0	0.101	0.093	0.086	0.080	0.074	0.068	0.063	0.058	0.053	0.049	0.045	0.041	0.038	0.034	0.031
Φ_1	0.072	0.071	0.070	0.069	0.068	0.067	0.065	0.063	0.062	0.060	0.059	0.057	0.056	0.054	0.052
Φ_2	0.048	0.048	0.047	0.047	0.047	0.046	0.046	0.046	0.046	0.045	0.044	0.044	0.043	0.043	0.042
Φ_3	0.035	0.035	0.036	0.036	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.034	0.034	0.034
Φ_4	0.028	0.028	0.028	0.028	0.028	0.029	0.029	0.029	0.029	0.029	0.028	0.028	0.028	0.028	0.028
Φ_5	0.023	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024	0.024
$1 - \beta(6)$	0.693	0.701	0.709	0.716	0.724	0.731	0.738	0.745	0.752	0.758	0.765	0.771	0.777	0.783	0.789

TABLE IV (continued)

Values of Φ_i and $1 - \beta(6)$ for soil thermal diffusivity of $0.5 \times 10^{-6} \text{ m}^2/\text{s}$

	Cable diameter or outer duct diameter (m)														
	0.180	0.185	0.190	0.195	0.200	0.205	0.210	0.215	0.220	0.225	0.230	0.235	0.240	0.245	0.250
Φ_0	0.029	0.026	0.024	0.022	0.020	0.018	0.016	0.015	0.013	0.012	0.011	0.010	0.009	0.008	0.007
Φ_1	0.050	0.048	0.047	0.045	0.043	0.042	0.040	0.038	0.037	0.035	0.033	0.032	0.030	0.029	0.027
Φ_2	0.041	0.041	0.040	0.039	0.038	0.037	0.037	0.036	0.035	0.034	0.033	0.033	0.032	0.031	0.030
Φ_3	0.034	0.033	0.033	0.032	0.032	0.032	0.031	0.031	0.031	0.030	0.030	0.029	0.029	0.028	0.028
Φ_4	0.028	0.028	0.027	0.027	0.027	0.027	0.027	0.027	0.026	0.026	0.026	0.025	0.025	0.025	0.025
Φ_5	0.024	0.024	0.024	0.024	0.024	0.023	0.023	0.023	0.023	0.023	0.023	0.023	0.022	0.022	0.022
$1 - \beta(6)$	0.794	0.800	0.805	0.811	0.816	0.821	0.826	0.830	0.835	0.840	0.844	0.848	0.853	0.857	0.861

APPENDIX A

BASIC THEORY FOR COMPUTATION OF CYCLIC RATING FACTORS
WITH ANY SHAPE OF LOAD CYCLE

The object of a cyclic rating factor calculation is to find, for a given shape of cycle, the factor M by which the steady-state rating (100% load factor) can be multiplied to give the peak current of the load cycle, when the peak cyclic conductor temperature will reach but not exceed the value used to determine the steady-state rating. A factor defined in this way has the steady-state temperature, which is usually the maximum permissible temperature, as its reference.

The daily current cycle is first expressed as a proportion of its maximum value so that the maximum value is unity. The cycle of cable heat-losses is then found by squaring the ordinates of the current cycle. The loss cycle is then decomposed into hourly rectangular pulses, each pulse magnitude being denoted by $Y_0, Y_1, Y_2, \dots, Y_{23}$, where Y_i is a measure of the squared current between i and $(i+1)$ hours prior to the expected time of the maximum conductor temperature. The latter is usually easily determined by inspection, but may not occur after the highest current values if these are of short duration only. The magnitude of Y_0 is therefore usually, but not necessarily, unity. Figure 1, page 34, shows a typical loss-load cycle treated in this way. The loss-load factor for this cycle μ is given by

$$\mu = \frac{1}{24} \sum_{i=0}^{23} Y_i \quad (19)$$

The temperature response of the cable to the cycle of losses can be found by regarding each change of magnitude as a separate step function which produces a transient response. The total response is the sum of all the transients.

Let the conductor temperature rise at time i hours after the application of a step function of losses corresponding to the rated current I_R be $\theta_R(i)$. Then the maximum conductor temperature rise above ambient for a daily cyclic current whose peak value is the rated current is:

$$\begin{aligned} \theta_m = & Y_0 \theta_R(1) + Y_1 [\theta_R(2) - \theta_R(1)] + Y_2 [\theta_R(3) - \theta_R(2)] + \dots \\ & \dots Y_5 [\theta_R(6) - \theta_R(5)] + \mu [\theta_R(\infty) - \theta_R(6)] \end{aligned} \quad (20)$$

It will be noted that detail of the load cycle is needed over a period of only 6 h before the time of maximum temperature, earlier values can be represented with sufficient accuracy by using an average. The loss-load factor μ provides this average.

$$M = \sqrt{\frac{\theta_R(\infty)}{\theta_m}} \quad (21)$$

where

$$M = \frac{1}{\left(\sum_{i=0}^5 Y_i \left[\frac{\theta_R(i+1)}{\theta_R(\infty)} - \frac{\theta_R(i)}{\theta_R(\infty)} \right] + \mu \left[1 - \frac{\theta_R(6)}{\theta_R(\infty)} \right] \right)^{1/2}} \quad (22)$$

$$\frac{\theta_R(i)}{\theta_R(\infty)} = \left[1 - k + k\beta(i) \right] \alpha(i) \quad (23)$$

$$\theta_R(0) = 0$$

Here $\beta(i)$ is the attainment factor for the cable or duct external surface i.e. the ratio of the temperature rise at time i to the rise in the steady state. $\alpha(i)$ is the attainment factor for the temperature rise of the conductor above the cable outside surface.

If the time periods being considered are sufficiently long compared with the thermal time constant of the cable, then $\alpha(i) = 1$. This is the condition for which the cable thermal capacitance may be neglected.

Putting $\alpha(i) = 1$ and substituting $[1 - k + k\beta(i)]$ for $\frac{\theta_R(i)}{\theta_R(\infty)}$ in equation (22) gives equation (2)

where:

$$B = Y_0 \Phi_0 + Y_1 \Phi_1 + \dots Y_5 \Phi_5$$

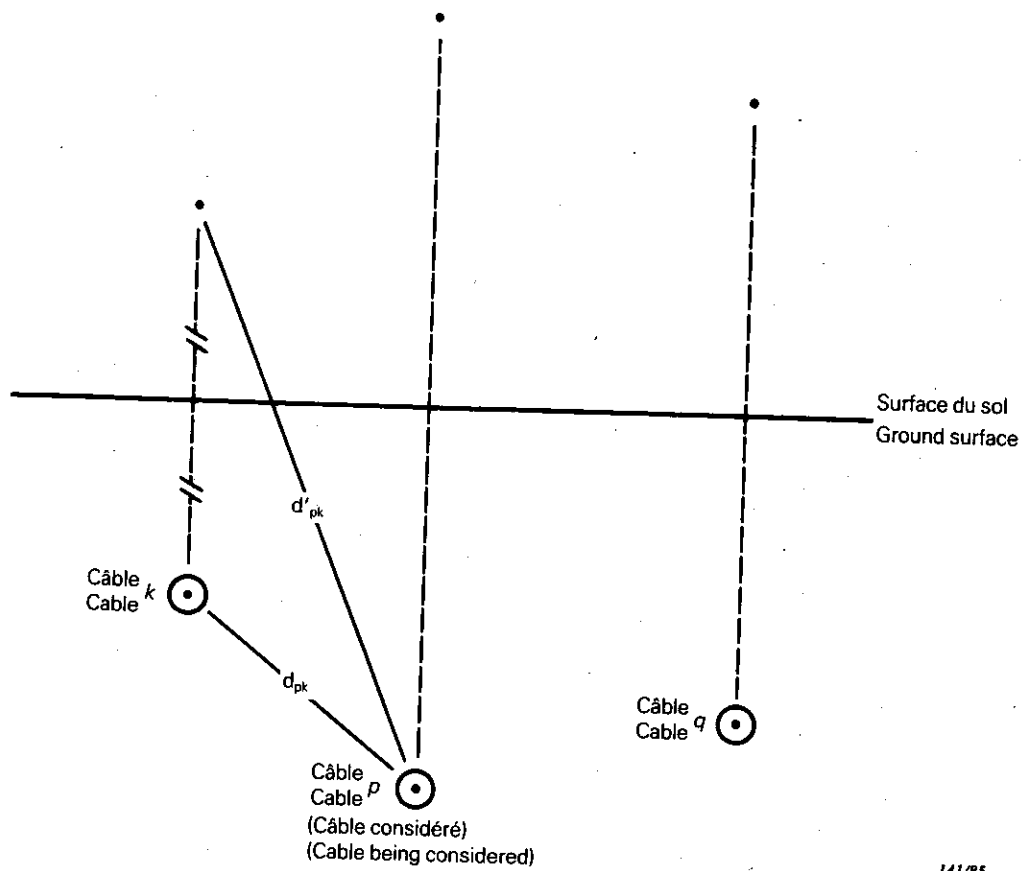
$$\text{and } \Phi_0 = \beta(1)$$

$$\Phi_i = \beta(i+1) - \beta(i) \text{ for } i = 1, 2 \dots 5 \quad (24)$$

The transient temperature rise of the cable or duct surface above ambient can be computed from:

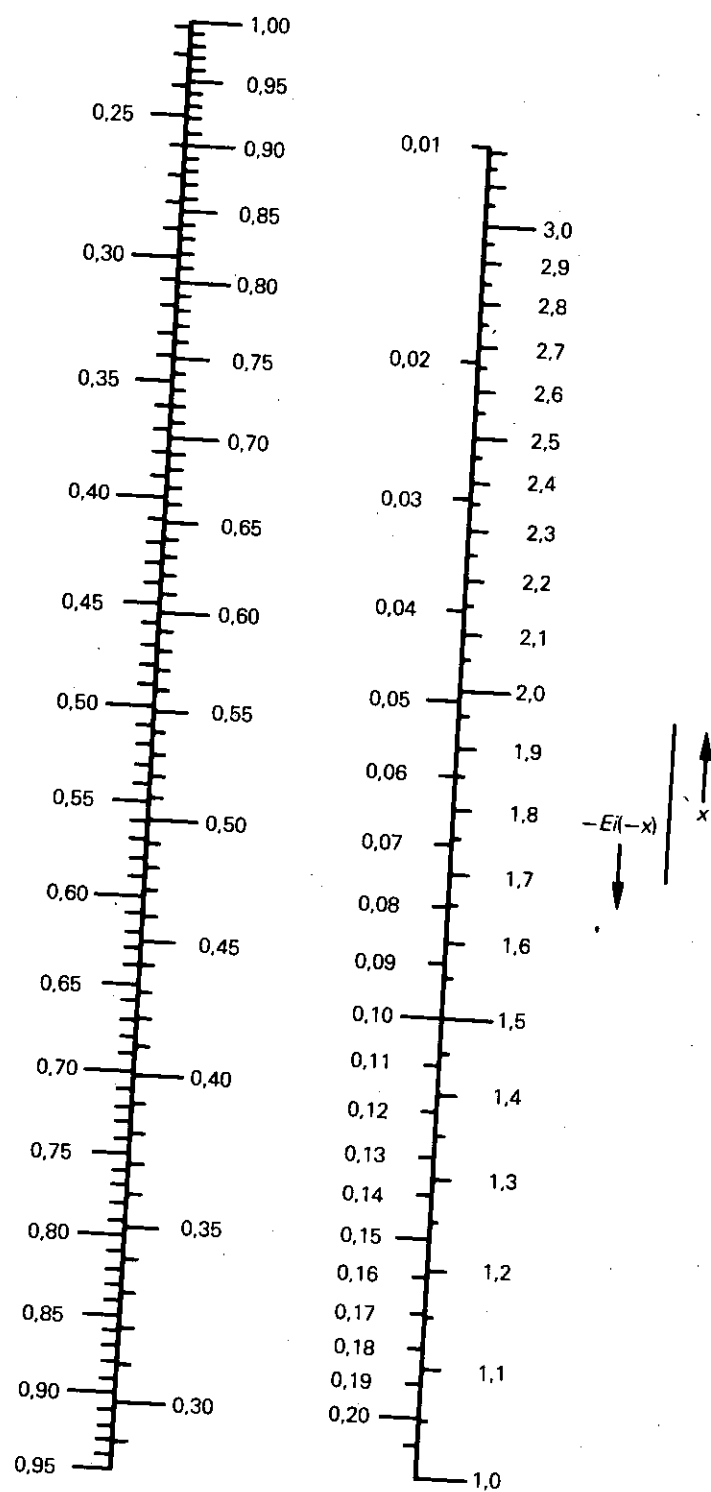
$$\beta(i) = \frac{-Ei \left(-\frac{D_c^2}{16t\delta} \right)}{2 \ln \left(\frac{4L}{D_c} \right)} \quad (25)$$

Φ_i and $[(1 - \beta(6))]$ can be tabulated against cable diameter, thus providing a rapid method for computing rating factors for single circuits where cable thermal capacitance can be neglected, for example see Table IV.



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FIG. 2. — Câbles ou circuits de câbles jointifs et leurs images.
Cables, or circuits of touching cables, and their images.



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FIG. 4. - Nomogramme de l'exponentielle intégrale $0,30 \leq x \leq 3,0$.
 Nomogram for exponential integral $0,30 \leq x \leq 3,0$.

IS TO FIGURES 3 AND 4 'NOMOGRAMS FOR EXPONENTIAL INTEGRAL'

The exponential integral $-Ei(-x)$ is defined in standard reference books, e.g. 'Handbook of Mathematical Functions' by M. Abramowitz and I. Stegun, pp. 228-231, from which the following rational approximations, adequate to give a result correct to four places of decimals, have been derived:

For $0 \leq x \leq 1$

$$-Ei(-x) = -\ln(x) + \sum_{i=0}^5 a_i x^i$$

0.5772
1.0000
0.2499
0.0552
0.0098
0.0011

For $1 < x < \infty$

$$-Ei(-x) = \frac{1}{x e^x} \left[\frac{x^2 + a_1 x + a_2}{x^2 + b_1 x + b_2} \right]$$

347
506

$b_1 = 3.3307$
 $b_2 = 1.6815$

The expressions given above are convenient for manual or machine use. However, the following single expression, based on one given in 'Tables of the Exponential Integral for Complex Arguments', National Bureau of Standards Applied Mathematics Series, 51, 1958, may be convenient for computer use although its evaluation time can be longer.

For all values of $x > 0$

$$-Ei(-x) = -0.5772 - \ln(x) - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n(n!)}$$

This is sufficient for the purposes of this standard if $-Ei(-x)$ is put equal to 0 for values of $x \geq 8$, and equal to $-0.5772 + x$ for $x < 0.01$.

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des isolés au papier imprégné sous gaine métallique pour des tensions assignées inférieures ou égales à 18/30 kV (avec âmes conductrices en cuivre ou aluminium et à l'exclusion des câbles à pression de gaz et à huile fluide).

-) Première partie: Essais.
-) Deuxième partie: Généralités et exigences de construction.
- 6) Câbles à huile fluide, à pression de gaz et de leurs dispositifs accessoires.
- 3) Première partie: Câbles au papier à huile fluide et à gaine métallique et accessoires pour des tensions alternatives inférieures ou égales à 400 kV.
- 3) Deuxième partie: Câbles à pression de gaz interne et accessoires pour des tensions alternatives inférieures ou égales à 275 kV. Modification n° 1 (1967).
- 3) Troisième partie: Câbles à pression de gaz externe (à compression de gaz) et accessoires pour des tensions alternatives inférieures ou égales à 275 kV. Modification n° 1 (1967).
- 4) Quatrième partie: Câbles à huile fluide en tuyau à isolation de papier imprégné sous forte pression d'huile et accessoires pour des tensions alternatives inférieures ou égales à 400 kV. Couleurs pour les conducteurs des câbles souples. Guide pour le choix des câbles à haute tension.
- Conducteurs et câbles isolés au polychlorure de vinyle, de tension nominale au plus égale à 450/750 V.
-) Première partie: Prescriptions générales.
-) Deuxième partie: Méthodes d'essais.
-) Troisième partie: Conducteurs pour installations fixes.
-) Quatrième partie: Câbles sous gaine pour installations fixes.
-) Cinquième partie: Câbles souples.
-) Sixième partie: Câbles pour ascenseurs et câbles pour connexions souples.
- Ames des câbles isolés.
-) Premier complément: Guide pour les limites dimensionnelles des âmes circulaires.
- Essais sur les gaines extérieures des câbles, qui ont une fonction spéciale de protection et sont appliquées par extrusion.
- Essais de choc des câbles et de leurs accessoires.
- Conducteurs et câbles isolés au caoutchouc, de tension nominale au plus égale à 450/750 V.
-) Première partie: Prescriptions générales.
-) Deuxième partie: Méthodes d'essais.
-) Troisième partie: Conducteurs isolés au silicone, résistant à la chaleur.
-) Quatrième partie: Câbles souples.
-) Cinquième partie: Câbles souples pour ascenseurs.
-) Sixième partie: Câbles souples pour électrodes de soudage à l'arc. Calcul du courant admissible dans les câbles en régime permanent (facteur de charge 100%).
- Caractéristiques des câbles électriques résistant au feu.
- Essais des câbles électriques soumis au feu.
- Première partie: Essai effectué sur un câble vertical.
- Troisième partie: Essais sur câbles en nappes. Modification n° 1 (1984).
- Câbles de transport d'énergie isolés par diélectriques massifs extrudés pour des tensions assignées de 1 kV à 30 kV. Modification n° 1 (1984).
- Méthodes d'essais pour les enveloppes isolantes et les gaines des câbles électriques rigides et souples (mélanges élastomères et thermoplastiques). Modification n° 1 (1984).
- Comparaison des câbles souples de la CEI et des câbles souples de l'Amérique du Nord.
- Câbles à isolant minéral de tension nominale ne dépassant pas 750 V.
- Calcul des valeurs minimales et maximales des dimensions extérieures moyennes des conducteurs et câbles à âmes circulaires en cuivre et de tension nominale au plus égale à 450/750 V.
- Guide aux limites de température de court-circuit des câbles électriques de tension assignée au plus égale à 0,6/1,0 kV.
- des gaz émis lors de la combustion des câbles électriques.
- Première partie: Détermination de la quantité de gaz acide halogéné émis lors de la combustion d'un matériau polymérisé prélevé sur un câble.
- Câbles chauffants de tension nominale 300/500 V pour le chauffage des locaux et la protection contre la formation de glace.
- des d'essais communes pour les matériaux d'isolation et de gainage des câbles électriques.
- Première partie: Méthodes d'application générale. Section deux: Méthodes de vieillissement thermique.
- Section trois: Méthode de détermination de la masse volumique. Essai d'absorption d'eau - Essai de rétraction.
- Section quatre: Essai à basse température.
- des capacités de transport des câbles pour les régimes de charge cycliques et de surcharge de secours.
- Première partie: Facteur de capacité de transport cyclique pour des câbles de tensions inférieures ou égales à 18/30(36) kV.

IEC publications prepared by Technical Committee No.20

- 55: — Paper-insulated metal-sheathed cables for rated voltages up to 18/30 kV (with copper or aluminium conductors and excluding gas-pressure and oil-filled cables).
- 55-1 (1978) Part 1: Tests.
- 55-2 (1981) Part 2: General and construction requirements.
- 141: — Tests on oil-filled and gas-pressure cables and their accessories.
- 141-1 (1976) Part 1: Oil-filled, paper-insulated, metal-sheathed cables and accessories for alternating voltages up to and including 400 kV.
- 141-2 (1963) Part 2: Internal gas-pressure cables and accessories for alternating voltages up to 275 kV. Amendment No. 1 (1967).
- 141-3 (1963) Part 3: External gas-pressure (gas compression) cables and accessories for alternating voltages up to 275 kV. Amendment No. 1 (1967).
- 141-4 (1980) Part 4: Oil-impregnated paper-insulated high pressure oil-filled pipe-type cables and accessories for alternating voltages up to and including 400 kV.
- 173 (1964) Colours of the cores of flexible cables and cords.
- 183 (1984) Guide to the selection of high-voltage cables.
- 227: — Polyvinyl chloride insulated cables of rated voltages up to and including 450/750 V.
- 227-1 (1979) Part 1: General requirements.
- 227-2 (1979) Part 2: Test methods.
- 227-3 (1979) Part 3: Non-sheathed cables for fixed wiring.
- 227-4 (1979) Part 4: Sheathed cables for fixed wiring.
- 227-5 (1979) Part 5: Flexible cables (cords).
- 227-6 (1981) Part 6: Lift cables and cables for flexible connections.
- 228 (1978) Conductors of insulated cables.
- 228A (1982) First supplement: Guide to the dimensional limits of circular conductors.
- 229 (1982) Tests on cable oversheaths which have a special protective function and are applied by extrusion.
- 230 (1966) Impulse tests on cables and their accessories.
- 245: — Rubber insulated cables of rated voltages up to and including 450/750 V.
- 245-1 (1985) Part 1: General requirements.
- 245-2 (1980) Part 2: Test methods.
- 245-3 (1980) Part 3: Heat resistant silicone insulated cables.
- 245-4 (1980) Part 4: Cords and flexible cables.
- 245-5 (1980) Part 5: Lift cables.
- 245-6 (1980) Part 6: Arc welding electrode cables.
- 287 (1982) Calculation of the continuous current rating of cables (100% load factor).
- 331 (1970) Fire-resisting characteristics of electric cables.
- 332: — Tests on electric cables under fire conditions.
- 332-1 (1979) Part 1: Test on a single vertical insulated wire or cable.
- 332-3 (1982) Part 3: Tests on bunched wires or cables. Amendment No. 1 (1984).
- 502 (1983) Extruded solid dielectric insulated power cables for rated voltages from 1 kV up to 30 kV. Amendment No. 1 (1984).
- 540 (1982) Test methods for insulation and sheaths of electric cables and cords (elastomeric and thermoplastic compounds). Amendment No. 1 (1984).
- 541 (1976) Comparative information on IEC and North-American flexible cord types.
- 702 (1981) Mineral insulated cables with a rated voltage not exceeding 750 V.
- 719 (1981) Calculation of the lower and upper limits for the average outer dimensions of cables with circular copper conductors and of rated voltages up to and including 450/750 V.
- 724 (1984) Guide to the short-circuit temperature limits of electric cables with a rated voltage not exceeding 0,6/1,0 kV.
- 754: — Test on gases evolved during combustion of electric cables.
- 754-1 (1982) Part 1: Determination of the amount of halogen acid gas evolved during the combustion of polymeric materials taken from cables.
- 800 (1984) Heating cables with a rated voltage of 300/500 V for comfort heating and prevention of ice formation.
- 811: — Common test methods for insulating and sheathing material of electric cables.
- 811-1-2 (1985) Part 1: Methods for general application. Section Two: Thermal ageing methods.
- 811-1-3 (1985) Section three: Method for determining the density.
- 811-1-4 (1985) Section Four: Tests at low temperatures.
- 853: — Calculation of the cyclic and emergency current rating of cables.
- 853-1 (1985) Part 1: Cyclic rating factor for cables up to and including 18/30(36) kV.